The higher inductive cumulative hierarchy satisfies constructive set-theoretic axioms

The Univalent Foundations Program

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Editorial note. Complete fixed-author source spans retained; original CC BY-SA attribution and macros reused. Supporting statements do not create extra counted problems.

Editorial note. Use bisimulation/identity and monic presentations to implement extensionality, replacement, set-indexed constructions and membership induction within the higher inductive cumulative hierarchy. Full author proof and native hierarchy constructors are retained.

Editorial note. The author labels the retained proof Sketch of proof. All nine author cases are included, with bisimulation and monic-presentation support; no elaboration has been written by the selector.

Allowed standard prerequisites (editorial).

- Basic univalent type theory. Identity elimination, function extensionality, univalence, dependent sums/products, propositional truncation, standard path transport and the explicitly supplied higher-inductive elimination rules. Standard elementary real/ordinal facts may be used; all target-specific local construction proofs are retained.

Source reference. Original reference 3.9.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 191–191. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 10.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1351–1351. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

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Source reference. Original reference 6.13: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1685–1685. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 7.2.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1797–1797. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

10.1 The cumulative hierarchy

We can define a cumulative hierarchy V of all sets in a given universe $\mathcal{U}$ as a higher inductive type, in such a way that V is again a set (in a larger universe $\mathcal{U}'$), equipped with a binary "membership" relation $x \in y$ which satisfies the usual laws of set theory.

Definition 10.1.1. The **cumulative hierarchy** V relative to a type universe $\mathcal{U}$ is the higher inductive type generated by the following constructors.

(i) For every $A : \mathcal{U}$ and $f : A \to V$, there is an element $\mathsf{set}(A, f) : V$.

(ii) For all $A, B : \mathcal{U}$, $f : A \to V$ and $g : B \to V$ such that

$$\big(\forall(a : A).\, \exists(b : B).\, f(a) =_V g(b)\big) \wedge \big(\forall(b : B).\, \exists(a : A).\, f(a) =_V g(b)\big) \tag{10.1.2}$$

there is a path $\mathsf{set}(A, f) =_V \mathsf{set}(B, g)$.

(iii) The 0-truncation constructor: for all $x, y : V$ and $p, q : x = y$, we have $p = q$.

In set-theoretic language, $\mathsf{set}(A, f)$ can be understood as the set (in the sense of classical set theory) that is the image of A under f, i.e. $\{\, f(a) \mid a \in A \,\}$. However, we will avoid this notation, since it would clash with our notation for subtypes (but see (10.1.3) and Definition 10.1.7 below).

The hierarchy V is bootstrapped from the empty map $\mathsf{rec}_{\mathbf{0}}(V) : \mathbf{0} \to V$, which gives the empty set as $\emptyset = \mathsf{set}(\mathbf{0}, \mathsf{rec}_{\mathbf{0}}(V))$. Then the singleton $\{\emptyset\}$ enters V through $\mathbf{1} \to V$, defined as $\star \mapsto \emptyset$, and so on. (The definition can also be adapted to include an arbitrary set of "atoms" or "urelements", by adding an additional point constructor.) The type V lives in the same universe as the base universe $\mathcal{U}$.

The second constructor of V has a form unlike any we have seen before: it involves not only paths in V (which in original reference 6.9 we claimed were slightly fishy) but truncations of sums of them. It certainly does not fit the general scheme described in original reference 6.13, and thus it may not be obvious what its induction principle should be. Fortunately, like our first definition of the 0-truncation in original reference 6.9, it can be re-expressed using auxiliary higher inductive types. We leave it to the reader to work out the details (see original reference 10.11).

At the end of the day, the induction principle for V (written in pattern matching language) says that given $P : V \to \mathsf{Set}$, in order to construct $h : \prod_{(x:V)} P(x)$, it suffices to give the following.

(i) For any $f : A \to V$, construct $h(\mathsf{set}(A, f))$, assuming as given $h(f(a))$ for all $a : A$.

(ii) Verify that if $f : A \to V$ and $g : B \to V$ satisfy (10.1.2), then $h(\mathsf{set}(A, f)) =^P_q h(\mathsf{set}(B, g))$, where q is the path arising from the second constructor of V and (10.1.2), assuming inductively that $h(f(a))$ and $h(g(b))$ are defined for all $a : A$ and $b : B$, and that the following condition holds:

$$\begin{aligned} &\big(\forall(a : A).\, \exists(b : B).\, \exists(p : f(a) = g(b)).\, h(f(a)) =^P_p h(g(b))\big) \\ \wedge\quad &\big(\forall(b : B).\, \exists(a : A).\, \exists(p : f(a) = g(b)).\, h(f(a)) =^P_p h(g(b))\big) \end{aligned}$$

The second clause checks that the map being defined must respect the paths introduced in (10.1.2). As usual when we state higher induction principles using pattern matching, it may seem tautologous, but is not. The point is that "$h(f(a))$" is essentially a formal symbol which we cannot peek inside of, which $h(\mathsf{set}(A, f))$ must be defined in terms of. Thus, in the second clause, we assume equality of these formal symbols when appropriate, and verify that the elements resulting from the construction of the first clause are also equal. Of course, if P is a family of mere propositions, then the second clause is automatic.

Observe that, by induction, for each $v : V$ there merely exist $A : \mathcal{U}$ and $f : A \to V$ such that $v = \mathsf{set}(A, f)$. Thus, it is reasonable to try to define the **membership relation** $x \in v$ on V by setting:

$$(x \in \mathsf{set}(A, f)) :\equiv (\exists(a : A).\, x = f(a)).$$

To see that the definition is valid, we must use the recursion principle of V. Thus, suppose we have a path $\mathsf{set}(A, f) = \mathsf{set}(B, g)$ constructed through (10.1.2). If $x \in \mathsf{set}(A, f)$ then there merely is $a : A$ such that $x = f(a)$, but by (10.1.2) there merely is $b : B$ such that $f(a) = g(b)$, hence $x = g(b)$ and $x \in \mathsf{set}(B, g)$. The converse is symmetric.

The **subset relation** $x \subseteq y$ is defined on V as usual by

$$(x \subseteq y) :\equiv \forall(z : V).\, z \in x \Rightarrow z \in y.$$

A **class** may be taken to be a mere predicate on V. We can say that a class $C : V \to \mathsf{Prop}$ is a V**-set** if there merely exists $v : V$ such that

$$\forall(x : V).\, C(x) \Leftrightarrow x \in v.$$

We may also use the conventional notation for classes, which matches our standard notation for subtypes:

$$\{\, x \mid C(x) \,\} :\equiv \lambda x.\, C(x). \tag{10.1.3}$$

A class $C : V \to \mathsf{Prop}$ will be called $\mathcal{U}$**-small** if all of its values $C(x)$ lie in $\mathcal{U}$, specifically $C : V \to \mathsf{Prop}_{\mathcal{U}}$. Since V lives in the same universe $\mathcal{U}'$ as does the base universe $\mathcal{U}$ from which it is built, the same is true for the identity types $v =_V w$ for any $v, w : V$. To obtain a well-behaved theory in the absence of propositional resizing, therefore, it will be convenient to have a $\mathcal{U}$-small "resizing" of the identity relation, which we can define by induction as follows.

Definition 10.1.4. Define the **bisimulation** relation

$$\sim : V \times V \longrightarrow \mathsf{Prop}_{\mathcal{U}}$$

by double induction over V, where for $\mathsf{set}(A, f)$ and $\mathsf{set}(B, g)$ we let:

$$\mathsf{set}(A, f) \sim \mathsf{set}(B, g) :\equiv \big(\forall(a : A).\, \exists(b : B).\, f(a) \sim g(b)\big) \wedge \big(\forall(b : B).\, \exists(a : A).\, f(a) \sim g(b)\big).$$

To verify that the definition is correct, we just need to check that it respects paths $\mathsf{set}(A, f) = \mathsf{set}(B, g)$ constructed through (10.1.2), but this is obvious, and that $\mathsf{Prop}_{\mathcal{U}}$ is a set, which it is. Note that $u \sim v$ is in $\mathsf{Prop}_{\mathcal{U}}$ by construction.

Lemma 10.1.5. *For any $u, v : V$ we have $(u =_V v) = (u \sim v)$.*

Proof. An easy induction shows that $\sim$ is reflexive, so by transport we have $(u =_V v) \to (u \sim v)$. Thus, it remains to show that $(u \sim v) \to (u =_V v)$. By induction on u and v, we may assume they are $\mathsf{set}(A, f)$ and $\mathsf{set}(B, g)$ respectively. (We can ignore the path constructors of V, since $(u \sim v) \to (u =_V v)$ is a mere proposition.) Then by definition, $\mathsf{set}(A, f) \sim \mathsf{set}(B, g)$ implies $(\forall(a : A).\, \exists(b : B).\, f(a) \sim g(b))$ and conversely. But the inductive hypothesis then tells us that $(\forall(a : A).\, \exists(b : B).\, f(a) = g(b))$ and conversely. So by the path constructor for V we have $\mathsf{set}(A, f) =_V \mathsf{set}(B, g)$. □

One might think that we could omit the 0-truncation constructor of V and *prove* that V is 0-truncated by applying original reference 7.2.2 to the bisimulation. However, in the proof of Lemma 10.1.5 we used the fact that V is 0-truncated, to conclude that $(u \sim v) \to (u =_V v)$ is a mere proposition so that in the induction it suffices to assume u and v are $\mathsf{set}(A, f)$ and $\mathsf{set}(B, g)$.

Now we can use the resized identity relation to get the following useful principle.

Lemma 10.1.6. *For every $u : V$ there is a given $A_u : \mathcal{U}$ and monic $m_u : A_u \rightarrowtail V$ such that $u = \mathsf{set}(A_u, m_u)$.*

Proof. Take any presentation $u = \mathsf{set}(A, f)$ and factor $f : A \to V$ as a surjection followed by an injection:

$$f = m_u \circ e_u : A \twoheadrightarrow A_u \rightarrowtail V.$$

Clearly $u = \mathsf{set}(A_u, m_u)$ if only A_u is still in $\mathcal{U}$, which holds if the kernel of $e_u : A \twoheadrightarrow A_u$ is in $\mathcal{U}$. But the kernel of $e_u : A \twoheadrightarrow A_u$ is the pullback along $f : A \to V$ of the identity on V, which we just showed to be $\mathcal{U}$-small, up to equivalence. Now, this construction of the pair (A_u, m_u) with $m_u : A_u \rightarrowtail V$ and $u = \mathsf{set}(A_u, m_u)$ from $u : V$ is unique up to equivalence over V, and hence up to identity by univalence. Thus by the principle of unique choice (3.9.2) there is a map $c : V \to \sum_{(A:\mathcal{U})}(A \to V)$ such that $c(u) = (A_u, m_u)$, with $m_u : A_u \rightarrowtail V$ and $u = \mathsf{set}(c(u))$, as claimed. □

Definition 10.1.7. For $u : V$, the just constructed monic presentation $m_u : A_u \rightarrowtail V$ such that $u = \mathsf{set}(A_u, m_u)$ may be called the **type of members** of u and denoted $m_u : [u] \rightarrowtail V$, or even $[u] \rightarrowtail V$. We can think of $[u]$ as the "subclass of V consisting of members of u".

Theorem 10.1.8. *The following hold for $(V, \in)$:*

(i) extensionality:

$$\forall(x, y : V).\, x \subseteq y \wedge y \subseteq x \Leftrightarrow x = y.$$

(ii) empty set: *for all* $x : V$*, we have* $\neg(x \in \emptyset)$.

(iii) pairing: *for all* $u, v : V$*, the class* $\{u, v\} :\equiv \{\, x \mid x = u \vee x = v \,\}$ *is a* V*-set.*

(iv) infinity: *there is a* $v : V$ *with* $\emptyset \in v$ *and* $x \in v$ *implies* $x \cup \{x\} \in v$.

(v) union: *for all* $v : V$*, the class* $\cup v :\equiv \{\, x \mid \exists(u : V).\, x \in u \in v \,\}$ *is a* V*-set.*

(vi) function set: *for all* $u, v : V$*, the class* $v^u :\equiv \{\, x \mid x : u \to v \,\}$ *is a* V*-set.*[1]

(vii) $\in$-induction: *if* $C : V \to \mathsf{Prop}$ *is a class such that* $C(a)$ *holds whenever* $C(x)$ *for all* $x \in a$*, then* $C(v)$ *for all* $v : V$.

(viii) replacement: *given any* $r : V \to V$ *and* $x : V$*, the class*

$$\{\, y \mid \exists(z : V).\, z \in x \wedge y = r(z) \,\}$$

is a V*-set.*

(ix) separation: *given any* $a : V$ *and* $\mathcal{U}$*-small* $C : V \to \mathsf{Prop}_{\mathcal{U}}$*, the class*

$$\{\, x \mid x \in a \wedge C(x) \,\}$$

is a V*-set.*

Sketch of proof.

(i) Extensionality: if $\mathsf{set}(A, f) \subseteq \mathsf{set}(B, g)$ then $f(a) \in \mathsf{set}(B, g)$ for every $a : A$, therefore for every $a : A$ there merely exists $b : B$ such that $f(a) = g(b)$. The assumption $\mathsf{set}(B, g) \subseteq \mathsf{set}(A, f)$ gives the other half of (10.1.2), therefore $\mathsf{set}(A, f) = \mathsf{set}(B, g)$.

(ii) Empty set: suppose $x \in \emptyset = \mathsf{set}(\mathbf{0}, \mathsf{rec}_{\mathbf{0}}(V))$. Then $\exists(a : \mathbf{0}).\, x = \mathsf{rec}_{\mathbf{0}}(V, a)$, which is absurd.

(iii) Pairing: given u and v, let $w = \mathsf{set}(\mathbf{2}, \mathsf{rec}_{\mathbf{2}}(V, u, v))$.

(iv) Infinity: take $w = \mathsf{set}(\mathbb{N}, I)$, where $I : \mathbb{N} \to V$ is given by the recursion $I(0) :\equiv \emptyset$ and $I(n + 1) :\equiv I(n) \cup \{I(n)\}$.

(v) Union: Take any $v : V$ and any presentation $f : A \to V$ with $v = \mathsf{set}(A, f)$. Then let $\tilde{A} :\equiv \sum_{(a:A)} [fa]$, where $m_{fa} : [fa] \rightarrowtail V$ is the type of members from Definition 10.1.7. $\tilde{A}$ is plainly $\mathcal{U}$-small, and we have $\cup v :\equiv \mathsf{set}(\tilde{A}, \lambda x.\, m_{f(\mathsf{pr}_1(x))}(\mathsf{pr}_2(x)))$.

(vi) Function set: given $u, v : V$, take the types of members $[u] \rightarrowtail V$ and $[v] \rightarrowtail V$, and the function type $[u] \to [v]$. We want to define a map

$$r : ([u] \to [v]) \longrightarrow V$$

with "$r(f) = \{\, (x, f(x)) \mid x : [u] \,\}$", but in order for this to make sense we must first define the ordered pair (x, y), and then we take the map $r' : x \mapsto (x, f(x))$, and then we can put $r(f) :\equiv \mathsf{set}([u], r')$. But the ordered pair can be defined in terms of unordered pairing as usual.

(vii) $\in$-induction: let $C : V \to \mathsf{Prop}$ be a class such that $C(a)$ holds whenever $C(x)$ for all $x \in a$, and take any $v = \mathsf{set}(B, g)$. To show that $C(v)$ by induction, assume that $C(g(b))$ for all $b : B$. For every $x \in v$ there merely exists some $b : B$ with $x = g(b)$, and so $C(x)$. Thus $C(v)$.

[1] Here $x : u \to v$ means that x is an appropriate set of ordered pairs, according to the usual way of encoding functions in set theory.

(viii) Replacement: let C denote the class in question. The statement "C is a V-set" is a mere proposition, so we may proceed by induction as follows. Supposing x is $\mathsf{set}(A, f)$, we claim that $w :\equiv \mathsf{set}(A, r \circ f)$ is the set we are looking for. If $C(y)$ then there merely exists $z : V$ and $a : A$ such that $z = f(a)$ and $y = r(z)$, therefore $y \in w$. Conversely, if $y \in w$ then there merely exists $a : A$ such that $y = r(f(a))$, so if we take $z :\equiv f(a)$ we see that $C(y)$ holds.

(ix) Let us say that a class $C : V \to \mathsf{Prop}$ is **separable** if for any $a : V$ the class

$$a \cap C :\equiv \{\, x \mid x \in a \wedge C(x) \,\}$$

is a V-set. We need to show that any $\mathcal{U}$-small $C : V \to \mathsf{Prop}_{\mathcal{U}}$ is separable. Indeed, given $a = \mathsf{set}(A, f)$, let $A' = \sum_{(x:A)} C(fx)$, and take $f' = f \circ i$, where $i : A' \to A$ is the obvious inclusion. Then we can take $a' = \mathsf{set}(A', f')$ and we have $x \in a \wedge C(x) \Leftrightarrow x \in a'$ as claimed. We needed the assumption that C lands in $\mathcal{U}$ in order for $A' = \sum_{(x:A)} C(fx)$ to be in $\mathcal{U}$. □